

AWS Infrastructure Cost Analysis Report

Executive Summary

Current Monthly Cost: \$120.82 USD

Annual Cost Projection: \$1,449.84 USD

Status: Pre-launch (0 users)

Billing Period: May 1-31, 2025

Current AWS Services Breakdown

1. Relational Database Service (RDS) - \$49.04 (40.6% of total)

Primary Cost Driver - Aurora MySQL Serverless: \$45.83 - 381.898 ACU-Hr at \$0.12 per hour - Designed for variable workloads - **Aurora Storage & I/O:** \$3.20 - 0.147 GB-Mo storage at \$0.10 per GB-month - 15,974,170 I/O requests at \$0.20 per million - **Backup Storage:** \$0.01 - 0.295 GB-Mo exceeding free tier

2. Elastic Compute Cloud (EC2) - \$38.16 (31.6% of total)

- **NAT Gateway:** \$33.48
 - 744 hours at \$0.045 per hour = \$33.48
 - 0.173 GB data processed
- **EC2 Instance (t3.nano):** \$3.87
 - 744 hours at \$0.0052 per hour
 - Linux/UNIX instance
- **EBS Storage:** \$0.80
 - 8 GB-Mo General Purpose SSD (gp2)

3. ElastiCache (Redis) - \$12.65 (10.5% of total)

- **T3 Micro Cache Node:** \$12.65
 - 744 hours at \$0.017 per hour
 - Redis caching service

4. Virtual Private Cloud (VPC) - \$7.44 (6.2% of total)

- **Public IPv4 Addresses:** \$7.44
 - 1,488 hours at \$0.005 per hour
 - Indicates 2 public IP addresses running continuously

5. AWS WAF (Web Application Firewall) - \$6.00 (5.0% of total)

- **Web ACL:** \$5.00 per month
- **Rules:** \$1.00 per rule per month
- **Request Processing:** \$0.00 (within free tier)

6. Route 53 (DNS) - \$1.00 (0.8% of total)

- **Hosted Zones:** \$1.00 (2 zones at \$0.50 each)
- **DNS Queries:** \$0.01 (15,814 queries, mostly free)

7. Other Services - \$6.53 (5.4% of total)

- **ECR (Container Registry):** \$0.10
- **API Gateway:** \$0.01
- **CloudFront:** \$0.00 (free tier)
- **CloudWatch:** \$0.00 (free tier)
- **Lambda:** \$0.00 (free tier)
- **SQS:** \$0.00 (free tier)
- **S3:** \$0.00 (free tier)
- **Data Transfer:** \$0.00 (free tier)

Annual Cost Projection (Current Infrastructure)

Service	Monthly Cost	Annual Cost
RDS Aurora	\$49.04	\$588.48
EC2 & EBS	\$38.16	\$457.92
ElastiCache	\$12.65	\$151.80
VPC	\$7.44	\$89.28
WAF	\$6.00	\$72.00
Route 53	\$1.00	\$12.00
Other Services	\$0.53	\$6.36
Total (Pre-Tax)	\$114.82	\$1,377.84
Tax	\$6.00	\$72.00
Grand Total	\$120.82	\$1,449.84

User Growth Cost Projections

Scenario 1: 100 Active Users

Estimated Monthly Cost: \$180-220 - **Database:** +\$30-50 (increased I/O and storage) - **Compute:** +\$20-30 (additional EC2 capacity) - **Data Transfer:** +\$10-15 - **Other Services:** +\$5-10

Scenario 2: 1,000 Active Users

Estimated Monthly Cost: \$400-600 - **Database:** +\$150-200 (higher ACU usage, more storage) - **Compute:** +\$80-120 (auto-scaling groups) - **Data Transfer:** +\$40-60 - **CloudFront:** +\$10-20 (CDN usage) - **Other Services:** +\$20-30

Scenario 3: 10,000 Active Users

Estimated Monthly Cost: \$1,200-2,000 - **Database:** +\$500-800 (multiple Aurora instances) - **Compute:** +\$400-600 (multiple AZs, larger instances) - **Data Transfer:** +\$150-250 - **CloudFront:** +\$50-100 - **Load Balancing:** +\$30-50 - **Monitoring & Logging:** +\$20-40

Architecture Assessment

Current Setup Strengths

- **Serverless Aurora:** Scales automatically with demand
- **Modern Architecture:** Microservices-ready with containers
- **Security:** WAF protection implemented
- **Caching:** Redis for performance optimization
- **Monitoring:** CloudWatch integration

Scaling Considerations

- **Database:** Aurora Serverless handles variable loads well
- **Compute:** Current t3.nano will need upgrading for production
- **Networking:** NAT Gateway provides high availability
- **Storage:** S3 integration for file storage scaling

Optimizations

Pre-Launch Phase

1. **Optimize NAT Gateway costs** for development environment
2. **Monitor Aurora ACU usage** patterns
3. **Implement cost alerts** at \$150/month threshold
4. **Review public IP requirements**

Post-Launch Phase

1. **Implement auto-scaling** for EC2 instances
2. **Add CloudFront CDN** for global performance
3. **Consider Reserved Instances** for predictable workloads
4. **Implement comprehensive monitoring** and alerting

Cost Management Strategy

1. **Monthly Reviews:** Track cost per user metrics
2. **Budgeting:** Set up AWS Budget alerts
3. **Reserved Capacity:** Consider 1-year commitments after 6 months
4. **Right-sizing:** Regular instance optimization reviews

Key Metrics to Track

- Cost per User per Month
- Database utilization and I/O patterns
- Compute resource utilization
- Data transfer patterns
- Cache hit ratios